

Time: Three Hours

Maximum Marks: 75

SECTION - A ($10 \times 2 = 20$ Marks)**Answer Any TEN Questions**

1. Find the L.C.M of 16, 24, 36 and 54.
2. Find the average of first 50 natural numbers.
3. Reduce $\frac{391}{667}$ to lowest terms.
4. Two numbers are in the ratio of 15: 11. If their H.C.F. is 13, find the numbers.
5. A man buys an article for Rs. 27.50 and sells it for Rs. 28.60. Find his gain percent.
6. Evaluate: 28 % of 450 + 45% of 280.
7. A can finish a work in 18 days and B can do the same work in half the time A takes. Then, working together what part of the same work they can finish in a day?
8. A cyclist covers a distance of 750 m in 2 min 30 sec. What is the cyclist's speed in km/hr?
9. Find the value $^{10}C_3$.
10. Three unbiased coins are tossed. What is the probability of getting at most two heads?

11. Find the present worth of Rs.930 due 3 years hence at 8 % annum.
Also, find the discount.
12. A man buys Rs. 25 shares in company which pays 9 % dividend. The money invested is such that it gives 10 % on investment. At what price did he buy the shares?

SECTION - B ($5 \times 5 = 25$ Marks)
Answer Any FIVE Questions

13. Evaluate $\sqrt{248 + \sqrt{51 + \sqrt{169}}}$.
14. A bag contains 50p, 25p, and 10p coins in the ratio 5: 9: 4, amounting to Rs. 206. The number of coins of each type.
15. Worker A takes 8 hours to do a job. Worker B takes 10 hours to do the same job. How long should it take both A and B, working together but independently, to do the same job?
16. The diagonal of a cube is $6\sqrt{3}$ cm. Find its volume and surface area.
17. Two dice are thrown together. What is the probability that the sum of the numbers on the two faces is divisible by 4 or 6?
18. At what time between 2 and 3 o'clock will the hands of a clock be together?
19. Find the cost of 96 shares of Rs.10 each at $\frac{3}{4}$ discount, brokerage being $\frac{1}{4}$ per share.

SECTION - C ($3 \times 10 = 30$ Marks)
Answer Any THREE Questions

20. Evaluate $\frac{1}{(\sqrt{9}-\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} - \frac{1}{(\sqrt{6}-\sqrt{5})} + \frac{1}{(\sqrt{5}-\sqrt{4})}$.
21. A shopkeeper sold a TV set for Rs.17940 with a discount of 8% and earned a profit of 19.6%. What would have been the percentage of profit earned if no discount was offered?
22. 45 men can complete a work in 16 days. Six days after they started working, 30 more joined them. How many days will they now take to complete the remaining work?
23. Two cards are drawn at random from a pack of 52 cards. What is the probability that either both are black or both are queens?
24. Which is the better investment $7\frac{1}{2}$ % stocks at 105 or $6\frac{1}{2}$ % stocks at 94?